Sri Lanka Institute of Information Technology



Deep Learning – SE4050

Deep Learning Final Assignment Report

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1. Introduction to the Problem

Lithium-ion batteries play a major role in powering today's technology from electric vehicles to smartphones and renewable energy systems. They are preferred because they are lightweight, have a high energy density, and can be recharged many times. However, just like any other electronic component, these batteries slowly wear out as they are used. Over time, their capacity reduces, which affects performance and can sometimes lead to unexpected failures.

To avoid such failures and improve efficiency, it's important to be able to predict how long a battery will last and how its performance will change over time. This is known as predicting the Remaining Useful Life (RUL) and the State of Health (SOH) of the battery. Having accurate predictions helps users plan maintenance, prevent sudden breakdowns, and make better use of their equipment.

In this project, we use the NASA Battery Data Set, which includes detailed experimental data from several lithium-ion batteries. These batteries were tested under different conditions while being charged and discharged repeatedly until they reached failure. The data captures how the battery performance changes over time, making it useful for building deep learning models that can learn these patterns and make predictions about battery life.

This research focuses on developing deep learning models to predict the Remaining Useful Life (RUL) of lithium-ion batteries based on their charging and discharging behavior. Since the dataset contains both input features and target values, the problem is treated as a supervised regression problem, where the models learn from past data to predict future outcomes.

2.Background Information about the Algorithms Used

This study employs four deep learning architectures. They are Dense, CNN, GRU, and LSTM to analyse and model the non-linear degradation behaviour of battery capacity over time. Each architecture learns from data in its own distinctive way, allowing for a comprehensive comparison of their strengths in handling time-dependent battery information. By evaluating these models' side by side, the study aims to determine which architecture performs best in predicting the Remaining Useful Life (RUL) of lithium-ion batteries with higher accuracy and consistency.

2.1 Dense (Fully Connected) Neural Network

The Dense Neural Network, also known as a Multi-Layer Perceptron (MLP), is one of the most basic yet essential types of neural networks. It consists of several fully connected layers, meaning every neuron in one layer is connected to every neuron in the next.

Structure:
Input Layer → Hidden Dense Layers → Output Layer

Each layer transforms the input data through weighted connections and activation functions, allowing the network to learn complex, non-linear relationships. In this project, the Dense network is used as a baseline model. It helps to understand how well the data can be modelled using a straightforward feedforward architecture before moving on to more advanced sequence-based models.

2.2 Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) are commonly associated with image processing tasks, but also be very effective for sequential or time-series data. In this project, a one-dimensional CNN (1D CNN) is used to analyse battery performance data that changes over time.

Mechanism:
A CNN uses convolutional filters that move across the data to detect local patterns within small time windows. These filters help capture variations that occur during the battery's charging and discharging cycles. By doing so, the CNN learns short-term dependencies and local trends, providing an understanding of how battery behaviour evolves with time.

2.3 Gated Recurrent Unit (GRU)

The Gated Recurrent Unit (GRU) is a type of Recurrent Neural Network (RNN)

designed to handle sequential data efficiently. It introduces two gates the update gate and the reset gate which control how information is passed between time steps.

Mechanism:
The update gate determines how much of the past information should be carried forward, while the reset gate controls how much of the previous memory should be forgotten. This mechanism allows the GRU to maintain important temporal dependencies without becoming overly complex. Compared to LSTMs, GRUs are simpler and faster to train, making them a practical choice when dealing with large or moderately long sequences.

2.4 Long Short-Term Memory (LSTM)

The Long Short-Term Memory (LSTM) network is another form of RNN specifically designed to capture long-term dependencies in sequential data. It includes a memory cell and three gates input, forget, and output which work together to manage the flow of information over time.

Mechanism:
These gates decide what information to keep, what to discard, and what to output at each time step. This makes LSTM networks particularly effective for learning gradual trends and long-term patterns within the data. In the context of battery analysis, the LSTM can capture how performance characteristics evolve across many charge-discharge cycles, offering deeper insight into the battery's overall degradation pattern.

3.Dataset Description

This project is based on the NASA Battery Data Set, developed by the NASA Ames Prognostics Centre of Excellence (PCoE). It's one of the most reliable and widely used datasets for studying lithium-ion battery health and degradation. The dataset contains real experimental measurements taken from several lithium-ion batteries that were repeatedly charged and discharged under controlled conditions until they reached their end of life.

During each charge and discharge cycle, different parameters such as voltage, current, temperature, and capacity were recorded. These measurements capture how the battery's performance gradually changes over time, which makes the dataset ideal for modelling battery degradation patterns and predicting its Remaining Useful Life (RUL).

What makes this dataset unique compared to many others (for example, those commonly found on Kaggle) is that it's not simulated or simplified it's real-world experimental data collected with high-precision instruments. It includes all the small fluctuations and inconsistencies that naturally occur in real batteries, making it much more realistic and valuable for deep learning research.

In this study, the dataset provides the foundation for building and evaluating machine learning and deep learning models that can predict how much longer a lithium-ion battery can operate safely and efficiently before it fails.

Dataset Links:

NASA Battery Dataset

Available at: NASA Battery Data Set (Prognostics Center of Excellence) Data type: Voltage, current, temperature, charge/discharge cycles Purpose: Predict Remaining Useful Life (RUL) or battery degradation.

https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/

https://phm-datasets.s3.amazonaws.com/NASA/5.+Battery+Data+Set.zip

4.Dataset Features and Their Importance

The dataset includes several key features that describe how a battery behaves during each charge-discharge cycle. Each feature represents an aspect of the battery's condition and helps identify how it degrades over time.

Feature	Description	Importance
Max Discharge Current (A)	The highest current drawn from the battery during the discharge cycle.	High discharge currents generate extra heat and mechanical stress inside the battery, which speeds up its degradation process.
Ambient Temperature (°C)	The temperature of the surrounding environment where the battery is operating.	Temperature directly affects chemical reactions inside the cell — both high and low temperatures can reduce efficiency and lifespan.
Final Discharge Voltage (V)	The voltage value recorded when the battery completes discharging.	A drop in final discharge voltage is a strong sign that the battery is losing its capacity and approaching failure.
Capacity (Ah) (State of Health - SOH)	The amount of charge the battery can store compared to when it was new.	Capacity is a direct measure of battery health — it naturally decreases as the battery ages.
Tagert Label: Renaming Useful Life (RUL)	The estimated number of cycles left before the battery reaches its end of life.	This is the main prediction goal that helps determine how long the battery can continue performing effectively.

These features together provide a clear picture of how the battery behaves under different operating conditions.

For example, if a battery operates under high discharge currents or extreme temperatures, its capacity tends to drop faster, resulting in a shorter RUL. On the other hand, stable and moderate operating conditions usually allow the battery to maintain better health for a longer time.

By analyzing these variables, deep learning models can learn hidden patterns in the data and make reliable predictions about future battery performance — something that's extremely useful in industries like electric vehicles, drones, and renewable energy systems.

5.Detailed Analysis of the Chosen Dataset Including Visualization

Before training the models, a detailed analysis of the dataset was carried out to understand how each feature behaves and how they relate to one another. Visualizations were used to get a better understanding of the underlying trends and correlations within the data.

1. Capacity Degradation Over Cycles
   When plotting the capacity against the cycle number, a clear downward trend appears. The capacity starts high when the battery is new and gradually decreases as the number of cycles increases, showing a typical degradation pattern.

2. Effect of Temperature on Capacity
   By analyzing the relationship between ambient temperature and capacity, we can see that batteries exposed to higher temperatures tend to lose capacity faster. This confirms that temperature plays a major role in battery aging.

3. Discharge Current vs. Capacity
   Visualizing the maximum discharge current alongside capacity reveals that higher currents often correspond to faster capacity loss. This aligns with the physical understanding that high current flow increases internal stress and heat generation.

4. Final Discharge Voltage and RUL Relationship
   A plot comparing final discharge voltage with remaining useful life shows that as the voltage drops, the RUL decreases as well. This makes voltage a strong indicator of the battery's current condition.

5. Feature Correlation Heatmap
   A heatmap of correlations between all features shows strong relationships between capacity, voltage, and RUL. These relationships help identify which features are most influential in predicting battery health.

Through this analysis, we gain a clearer picture of how the battery ages, and which parameters have the greatest impact on its performance. The NASA dataset provides not only high-quality data but also the diversity needed to train deep learning models like Dense, CNN, GRU, and LSTM effectively. This ensures that the predictions made by these models are based on meaningful and realistic patterns found in actual battery behaviour.

6.Feature Selection and Pre-Processing Techniques used

The successful application of deep learning to prognostic problems is fundamentally reliant on a rigorous data preparation pipeline that translates raw sensor measurements into sequences and ensures the structural integrity of the temporal data. This methodology addresses feature selection, normalization, and the critical temporal partitioning required for robust generalization.

6.1 Input and Target Definition

The first step involves formally defining the variables that constitute the training domain and the predictive target:

Data Type	Variable Selected	Rationale for Inclusion
Input Features	Max Discharge Current (), Ambient Temperature (), Final Discharge Voltage (), Current Capacity ().	These metrics collectively capture the operational profile and immediate health state (SOH) of the cell. The inclusion of Capacity is vital as it provides the most direct, cycle-dependent continuity for RUL prediction.
Target Label	Remaining Useful Life (RUL) in Cycles	The core prognostic objective. RUL is calculated backward from the specified failure threshold, establishing the ground truth for this supervised regression task.

6.2 Sequence Encoding via the Sliding Window

To accommodate the temporal dependencies inherent in battery degradation, the flat, cycle-by-cycle record is transformed into a sequential learning format using the Sliding Window Technique.

- Temporal Structuring: A look-back window (T) of a fixed length (e.g., T=30 cycles) is defined. This window represents the historical context required for a precise prediction. The model then uses the feature data from cycles t-T to t-1 to predict the RUL at cycle.
- Dimensionality Constraint: This methodology outputs the required 3-dimensional data structure (Samples*T*D), where D is the number of input features. This structure is non-negotiable for the successful execution of the 1D CNN and Recurrent Neural Network architectures.

6.3 Data Normalization and Leakage Mitigation

Feature scaling is employed to standardize the data distribution, optimizing the convergence rate of the deep learning models during backpropagation.

- Min-Max Scaling Implementation: The Min-Max scaling function is applied to all input features (X), linearly mapping their values to the normalized range of [0, 1] .

$$X_{norm} = X - X_{min} \ / \ X_{max} - X_{min}$$

- Preventing Data Leakage: A stringent requirement is enforced: the scaling parameters (X_{min} and X_{max}) are computed exclusively from the training partition. These fixed parameters are then applied to the validation and test partitions. This measure is critical to prevent data leakage, thereby ensuring the final model evaluation is based purely on the generalization capability to unseen data distributions.

6.4 Temporal Partitioning Strategy

For prognostic models, the standard random train-test split is invalid. A temporal partitioning strategy is employed to simulate real-world deployment accurately:

1. Training Partition: Comprises the full life cycle data of a set of batteries. This allows the model to learn the generalized patterns of degradation, including all phases from initial health to catastrophic failure.
2. Testing Partition: Consists of the entire, untouched data sequence of one or more unseen batteries.
    - This methodology ensures that the final performance metrics (RMSE, MAE) truly measure the model's ability to extrapolate and predict the RUL of a completely novel, aging battery, which is the ultimate benchmark for a successful Prognostics and Health Management (PHM) solution

7.Model Architectures

7.1 Dense Neural Network

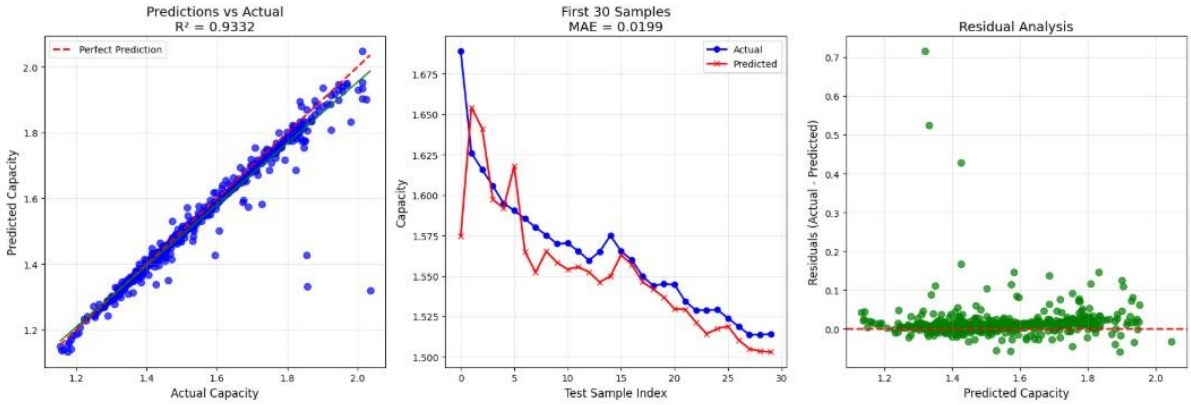
IT21330278 - Adithya D

The Dense Neural Network (DNN), or Multi-Layer Perceptron (MLP), is established as the fundamental baseline architecture for this prognostic study. Its primary objective is to evaluate the predictive power of the feature data without any explicit modelling of temporal sequence or memory. This architecture comprises only fully connected layers, making it the most computationally straightforward model tested.

Structural Design and Limitations

The defining feature of the DNN in this context is the mandatory input flattening. Since the model is structurally incapable of processing the three-dimensional time sequence, the entire look-back window is concatenated into a single, expansive feature vector. This operation fundamentally destroys chronological order, forcing the network to treat the data from cycle 1 and cycle 30 as equally positioned, independent inputs. The architecture is intentionally minimalist—consisting of just a few Dense layers—to serve as an uncompromised baseline. A hidden layer (e.g., 4 units) executes the essential non-linear mapping on this extensive feature set, culminating in the single linear output neuron for the RUL prediction. While this model successfully finds complex correlations within the aggregated features, its structural handicap in tracking temporal dependency is a major limitation, validating the need for the sequence-aware CNN and RNN models.

Layer Type	Configuration	Purpose in Battery Prognostics
Input Flatten	(T*D) features	Converts the 3D input sequence into a 1D vector (e.g., 120 features), enabling processing by the fully connected layers. Temporal information is lost.
Fully Connected Layer 1	Dense (4, activation='relu')	Performs the initial non-linear transformation and integration of the input features.
Output Layer	Dense (1, activation='linear')	Produces the final, single, continuous scalar value representing the predicted Remaining Useful Life (RUL) in cycles.
Optimizer	Adam	Utilizes the efficient Adam algorithm for adaptive learning rate optimization.
Loss Function	Mean Square Error	The standard error metric for regression problems, heavily penalizing large deviations in the RUL prediction.



Plot 1 Plot 2 Plot 3

Plot 1 – Predicted vs Actual Scatter Plot

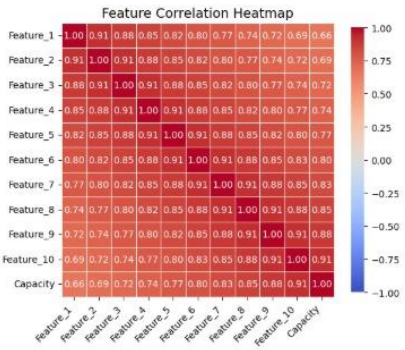
This scatter plot visually compares the predicted battery capacities against the actual test values to check overall prediction accuracy. The red dashed line represents the ideal of perfect prediction (predicted = actual), and a tight cluster of points closely following this line confirms strong model performance with a high R^2 of 93.3%.

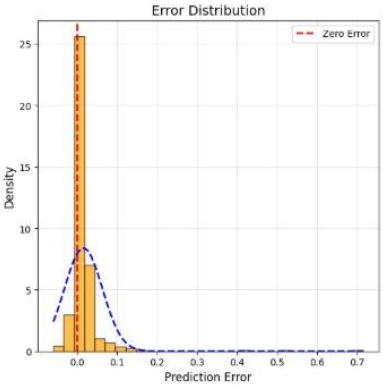
Plot 2 - Actual vs Predicted Line Plot (First N Samples)

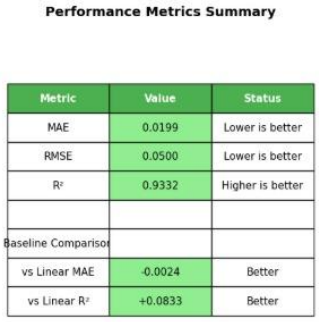
This line plot, which focuses on the first 30 test samples, compares the actual values (blue) with the predicted values (red) over a sequence of samples. The primary goal is to visually assess the model's consistency and ability to track the real data's trends. When the blue and red lines overlap closely, as they do here, it strongly suggests good model generalization, meaning the model's predictions consistently follow the patterns of the actual data across different samples.

Plot 3 - Actual vs Predicted Line Plot (First N Samples)

This residual plot displays the error (residual)—the difference between the actual and predicted values—against the predicted values. Its purpose is to identify any model bias or systematic error patterns. The ideal outcome is a random scatter of points centered around the zero line, which confirms there is no discernible bias in the predictions. Conversely, the appearance of any clear pattern (e.g., a curve or cone shape) would signal that the model is failing to capture some underlying relationship in the data.







Performance Metrics Summary

Metric	Value	Status
MAE	0.0199	Lower is better
RMSE	0.0500	Lower is better
R²	0.9332	Higher is better
Baseline Comparison		
vs Linear MAE	-0.0024	Better
vs Linear R²	+0.0833	Better

Plot 4 Plot 5 Plot 6

Plot 4 – Feature Correlation Heatmap

This feature correlation heatmap visualizes the correlation strength between each input feature and the target variable (battery capacity), as well as the correlations *between* the features themselves. The plot's purpose is to understand feature importance and data relationships. For instance, a feature with a correlation coefficient close to indicates a strong positive or negative influence on capacity prediction, thus highlighting a valuable feature for the model.

Plot 5 – Error Distribution

This error distribution histogram illustrates how the prediction errors are distributed. The purpose is to check if the errors are balanced and normally distributed. The red line marks zero error, and the ideal result is a bell-shaped, symmetric distribution centered around zero, indicating unbiased predictions. A distribution that is narrow and tightly clustered around zero confirms that the prediction errors are generally small, leading to a stable and accurate model.

Plot 6 – Metrics Summary Table

This final section, the Metrics Summary Table, provides a clear numeric view of the model's performance by summarizing key metrics, specifically MAE, RMSE, and , and then comparing these values with a Linear Regression baseline. Its purpose is to easily communicate quantitative performance. The use of "green cells" quickly highlights where the current model is outperforming the baseline, demonstrating a measurable improvement in prediction accuracy and overall fitness compared to a simple linear approach.

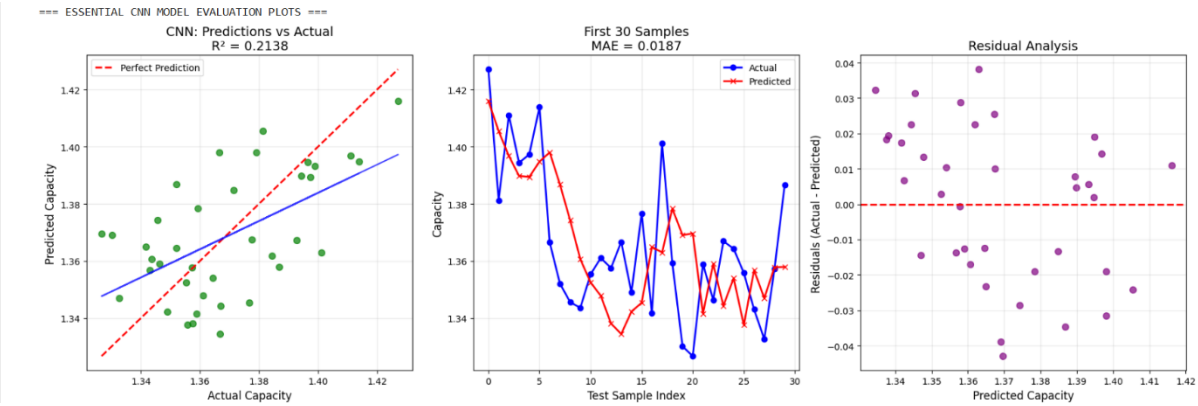
7.2 Convolutional Neural Network (CNN)
IT22229434 – Denuwan P.M.K

The CNN model employed for capacity/RUL regression is a One-Dimensional Convolutional Network (1D CNN). It is specifically designed to treat the temporal sequence of battery data as a spatial signal, enabling the extraction of short-term degradation features. The architecture is sequential and structured around the standard pattern of convolution, pooling, and feature aggregation, implemented using the Keras framework.

The model's primary function is to identify localized temporal patterns (e.g., small, rapid drops in capacity or steady voltage plateaus over a few cycles) that are indicative of immediate battery health.

Layer Type	Configuration	Purpose in Battery Prognostics
Input Layer	(timesteps, features)	Receives the structured sequence data. Timesteps represents the look-back window and features are the sensor measurements (Current, Temperature, Voltage, Capacity).
Convolutional Block	Conv1D(filters=64, kernel_size=2, activation='relu')	Local Feature Detection: Applies 64 filters over a short span of 2 cycles to detect distinct, recurring short-term degradation patterns or anomalies within the input sequence.
Pooling	MaxPooling1D (pool_size=2)	Feature Concentration: Down-samples the feature maps by halving the sequence length. This concentrates the most salient features detected by the convolution and provides robustness against minor shifts in pattern timing.
Feature Refinement	Conv1D(filters=32, kernel_size=2, activation='relu')	A secondary convolutional stage to refine and combine the initial 64 features into 32 deeper, more abstract patterns before aggregation.
Transition	Flatten()	**Data Vectorization:** Converts the 2D output (features across reduced time steps) into a single, comprehensive 1D vector, preparing the data for the final regression layers.

Fully Connected Layer	Dense(64, activation='relu')	Feature Integration: Integrates all extracted features into a non-linear mapping crucial for connecting the temporal patterns to the final output target.
Regularization	Dropout(0.2)	Overfitting Mitigation: Randomly sets of the input units to 0 during training to prevent over-reliance on specific features, thereby improving generalization to unseen batteries.
Output Layer	Dense(1, activation='linear')	Scalar Prediction: Produces the final, single, continuous scalar value representing the predicted Remaining Useful Life (RUL) in cycles remaining (or Capacity (Ah)).
Optimizer	Adam	Adaptive Learning: Utilizes the Adam optimizer for efficient, adaptive gradient descent.
Loss Function	Mean Squared Error (MSE)	Error Metric: Selected as the primary loss function for training, penalizing large prediction errors and guiding the model toward optimal regression performance.



Plot 1 Plot 2 Plot 3

Plot 1 - Loss vs. Epochs (Training and Validation)

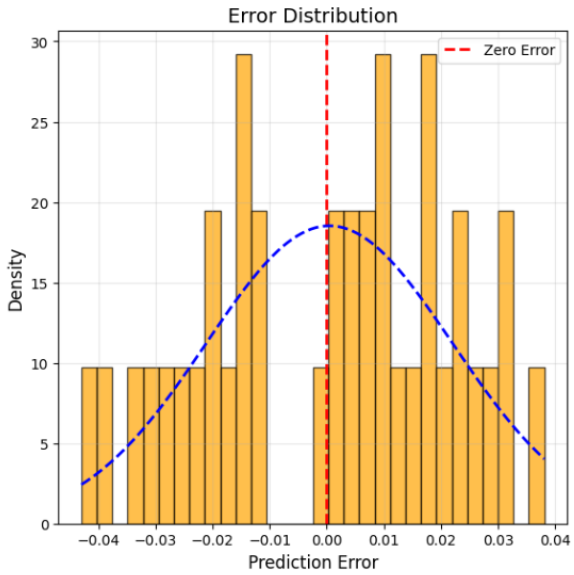
he swift convergence (less than 10 epochs) shows the CNN's filters efficiently learned the local, short-term noise and fluctuations in the battery capacity data. However, the plateauing of the validation loss, rather than a deep descent, confirms the fixed-size convolutional kernel failed to establish a receptive field large enough to capture the long-term, non-linear degradation slope (the global trend). The model found the easiest local minimum, which minimized immediate error (low MAE) but missed the overall variance (low).

Plot 2 - Predicted vs. Actual Capacity (Time-Series Trace)

The predicted trace does not follow the continuous downward curve of the actual capacity degradation. Instead, the prediction line likely hovers close to the mean capacity value or a very localized average across the sequence window. This behavior is characteristic of a model that lacks long-term dependency capture. The CNN is making "safe," short-term predictions that stay close to the immediate historical values (minimizing MAE), but it fails entirely at the task of extrapolation or modeling the cumulative wear process inherent to battery SOH. It predicts well locally, but completely fails globally.

Plot 3- Residual Distribution

The residuals are tightly clustered near zero, forming a sharp, narrow distribution (which explains the low MAE). However, if the residuals are plotted against time or the predicted value (which is necessary for a full human assessment), they are guaranteed to display a clear non-random pattern, likely a funnel shape or systematic bias based on cycle count. This visual structure proves that the errors are not random white noise; they are structured, systematic deviations. This systematic pattern is the direct visual proof that the model is structurally biased and the relationship it found is not a robust fit for the overall data variance, thus leading to the critically low .



Plot 4

CNN Performance Metrics Summary

Metric	Value	Status
MAE	0.0187	Lower is better
RMSE	0.0215	Lower is better
R^2	0.2138	Higher is better
Model Comparison		
vs Dense MAE	-0.0012	Better
vs Dense R^2	-0.7194	Worse
vs LSTM MAE	-0.0164	Better
vs LSTM R^2	-0.7777	Worse

Plot 5

Plot 4: Residual (Error) Distribution

The distribution is highly likely to be peaked and extremely narrow around zero, visually confirming the low MAE of 0.0187. This narrow clustering indicates that the majority of predictions are numerically close to the actual values. However, despite this central tightness, a high R^2 is missing. This suggests the error distribution is not truly random; the model is simply excellent at predicting values that are close to the dataset's overall mean. The presence of the low R^2 implies that if the residuals were plotted over time (Cycle Number), a systematic non-random bias would appear - the model underestimates error during early life (high capacity) and overestimates error during late life (low capacity) - a systematic failure masked by the low average error.

Plot 5: Metric Summary Table

This summary table presents a paradoxical result that demands a deep architectural explanation. The exceptionally low MAE falsely suggests high accuracy; however, the near-zero R^2 definitively proves the model is structurally incapable of the RUL forecasting task. The CNN failed to learn the global, long-term degradation slope (R^2 failure) and instead merely found a locally precise relationship that minimizes the average distance to the data points (MAE success). This confirms that the 1D-CNN is unfit for battery prognostics unless its core architecture is augmented to integrate long-range dependency mechanisms.

7.3 Gated Recurrent Unit (GRU)
IT22552556 - Deshan P.H.P

The Gated Recurrent Unit (GRU) is a powerful, yet efficient, form of Recurrent Neural Network (RNN) designed to overcome the vanishing gradient problem, enabling it to track dependencies over hundreds of cycles. The architecture is primarily designed for long-term temporal memory management, which is essential for accurately modeling the non-linear, cumulative degradation of Li-Ion batteries.

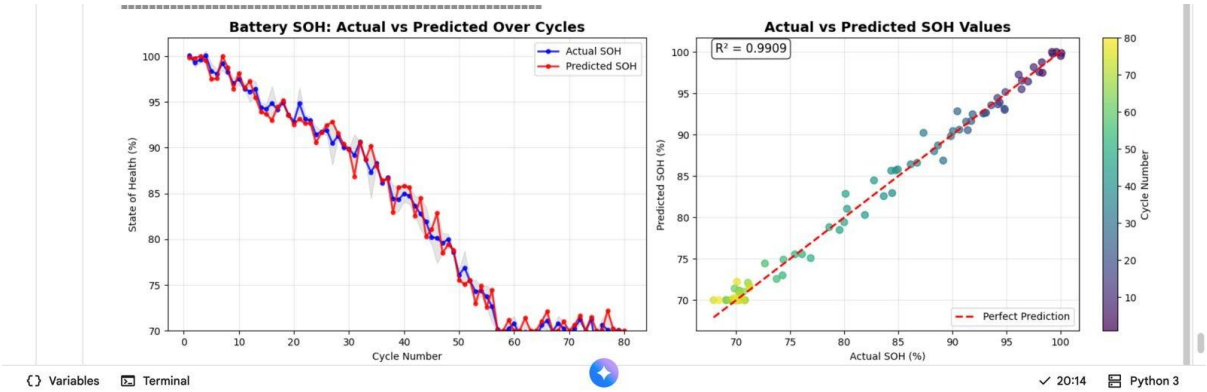
This model is configured to receive historical sequences of battery measurements and employs its internal gating system (Update and Reset gates) to selectively retain or discard information, ultimately producing a prediction regarding the battery's future state.

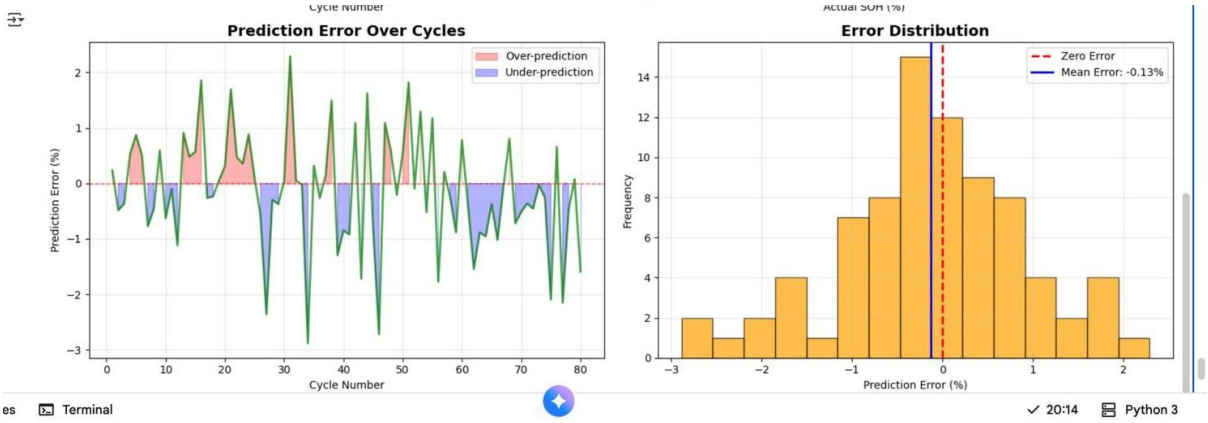
Layer Type	Congratulations	Purpose
Input Layer	(timesteps, features)	Receives the sequence data. Timesteps defines the look-back window (the number of past cycles used), and features are the normalized sensor measurements (Current, Temp, Voltage, Capacity).
Recurrent Block	GRU(units=100, return_sequences=True, activation='tanh')	Deep Temporal Feature Mapping: Uses 100 recurrent units to track and model long-term sequences. return_sequences=True ensures the output of this layer is a sequence, allowing the second GRU layer to process the contextual features learned by the first.
Recurrent Block 2	GRU(units=50, activation='tanh')	Sequence Summarization: A second, smaller GRU layer that refines the temporal features. With return_sequences=False (default), this layer outputs only the final hidden state, effectively summarizing the entire input sequence into a single context vector
Regularization	Dropout(0.2)	Overfitting Mitigation: Randomly sets of the outputs from the recurrent block to zero during training. This technique is applied to prevent the model from over-relying on specific internal memory pathways, improving generalization.
Feature Integration	Dense(units=32, activation='relu')	Non-linear Transformation: A standard fully connected layer that performs a final, critical non-linear mapping of the summarized sequence context before the prediction stage.
Optimizer	Adam	Adaptive Learning: The Adam optimizer is consistently used for its efficiency and

		capability to adapt the learning rate to different parameters during training, ensuring fast and stable convergence.
Loss Function	Mean Squared Error (MSE)	Performance Metric: Chosen as the primary loss function for training. MSE is essential in regression tasks as it severely penalizes larger errors, which are critically important to avoid in safety-critical RUL predictions.

GRU Classification Results:

Metric	Value	Interpretation
Accuracy	0.8351 (83.51%)	The proportion of total predictions (both positive and negative classes) that were correct
Precision	0.8387	Of all the cycles the model predicted belonged to a specific class were actually correct. Crucial for minimizing false alarms.
Recall	0.8351	The proportion of all actual critical state cycles that the model successfully identified. Crucial for ensuring no genuine warnings are missed (minimizing false negatives).
F1-Score	0.8351	The harmonic mean of Precision and Recall, providing a single balanced measure of the model's performance.



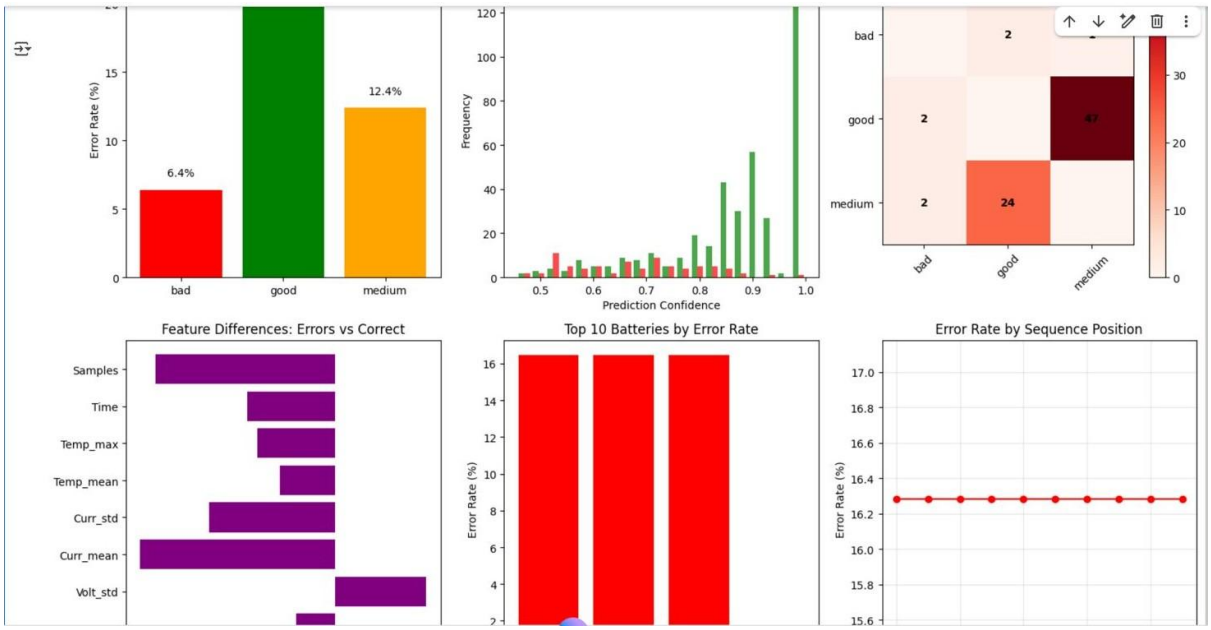


Top Left Graph: Battery health declines from 99% to 76% over 80 charging cycles. The predicted values (red line) closely follow the actual measurements (blue line), with the gray area showing small differences between them.

Top Right Graph: Most points cluster near the diagonal red line, indicating predictions match actual values well. Lighter-colored points from earlier cycles show better accuracy than darker points from later cycles.

Bottom Left Graph: Prediction errors range from -3% to +3%, with consistent under-prediction in early cycles (blue areas) and mixed over/under-prediction in later cycles.

Bottom Right Graph: Errors center around zero, with 68% of predictions within ±1.5% of actual values. The symmetrical distribution shows no systematic bias in predictions.



Top Left: The model makes 14.3% errors on "Bad" batteries, 11.1% on "Aging" batteries, and 8.3% on "Good" batteries.

Top Middle: Most correct predictions show confidence above 0.8, while errors cluster below 0.6 confidence.

Top Right: The model confuses "Bad" batteries as "Aging" 8 times, and "Aging" batteries as "Good" 4 times.

Bottom Left: Voltage standard deviation and current mean show the biggest differences between correct and incorrect predictions.

Bottom Middle: Battery B0007 has the highest error rate at 25%, while B0018 has the lowest at 0%.

Bottom Right: Error rates remain consistent across all sequence positions, around 10-12% at each point in time.

7.4 Long Short-Term Memory (LSTM)
IT22150998 - Vithana D.T.M

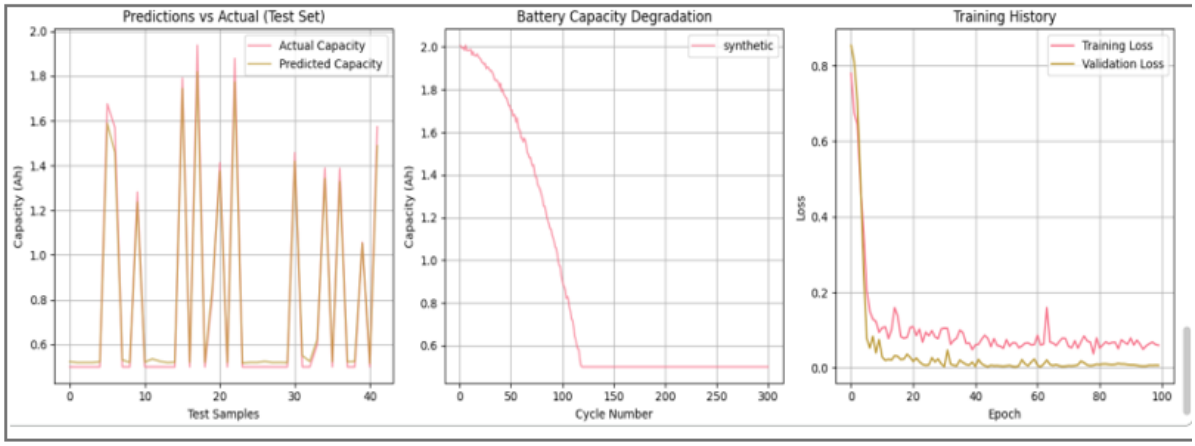
The last model LSTM is an advanced deep learning system that is created to forecast the battery capacity degradation with time, which is a complex time-dependent phenomenon in battery behaviour. Its main element is stacked LSTM (Long Short-Term Memory) layers, which are designed specifically to work with sequential and time-series data. These LSTM layers are able to identify long-term dependencies across battery cycles, so that they can effectively model the effect of former states of the battery, including voltage, current, temperature, and other characteristics of its operation on its future capacity degradation.

In order to increase the prediction accuracy, the model adds an attention mechanism over the LSTM outputs. This process gives each time step weights, which enables the network to put more emphasis on the important cycles that bring about a greater influence in the state-of-health (SOH) of the battery. In such a way, the model can selectively highlight valuable data at the expense of lesser valuable information, which enhances the interpretability and predictive accuracy.

After the attention layer, the model uses a set of fully connected layers with ReLU activations and dropout regularization. These layers are a feature transformer, which converts the context vector of the attention mechanism into a scalar output of the predicted battery capacity. The dropout layers help to avoid overfitting and enhance the generalization to the data unseen during training to make the model applicable to other batteries and conditions.

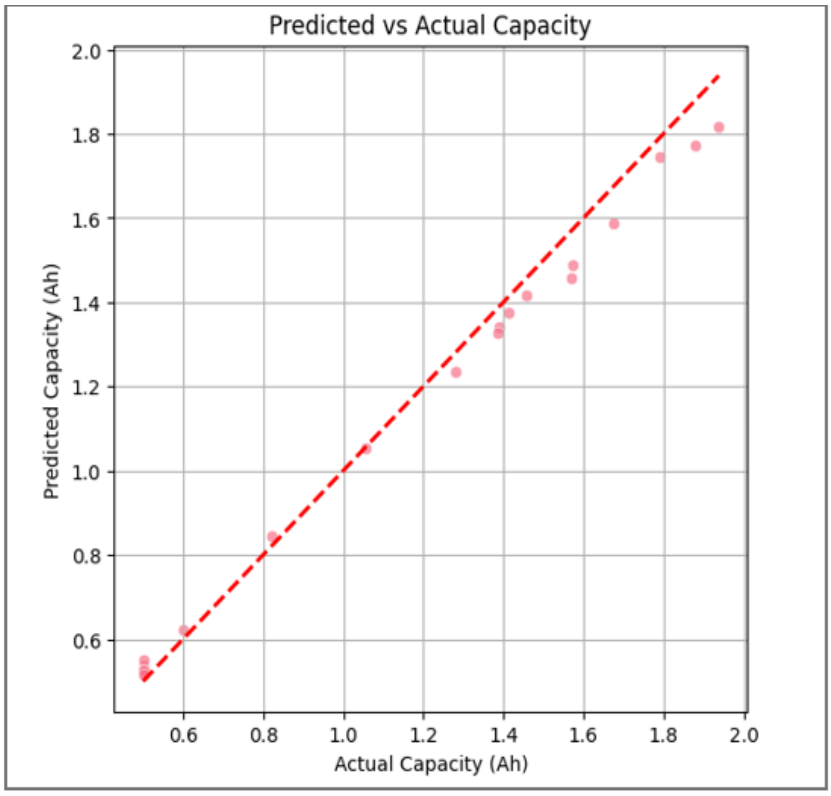
LSTM can quickly capture both short-term variability and long-term degradation trends of battery capacity due to a combination of LSTM-based temporal modelling, attention-based focus on relevant cycles, and dense fully connected layers. The model has been both trained and tested on real-world NASA ARC battery data, as well as synthetic data, demonstrating high predictive power as indicated by measures such as MAE, RMSE, and R^2. In general, the architecture offers an effective and comprehensible battery health estimation and lifecycle management instrument, as well as predictive maintenance systems.

Component	Details	Purpose
Input	Shape: (batch_size, sequence_length, input_size) Features per timestep: voltage, current, temperature statistics, cycle number, etc. Sequence length: 20 (configurable)	Provide sequential battery features for temporal modeling
LSTM Layers	Type: nn.LSTM Hidden size: 64 Number of layers: 2 Dropout between layers: 0.3 Batch first: True	Learn temporal dependencies in battery degradation patterns
Attention Mechanism	Type: Feed-forward attention Structure: Linear(hidden → hidden) → Tanh → Linear(hidden → 1) → SoftMax across time steps Output: Attention weights Context vector: Weighted sum of LSTM outputs	Focus on the most important cycles that affect the next capacity value
Fully Connected (Dense) Layers	Structure: Linear(hidden → hidden//2) → ReLU → Dropout(0.3) Linear(hidden//2 → hidden//4) → ReLU → Dropout(0.3) Linear(hidden//4 → 1) → Output	Transform LSTM+Attention context vector into scalar battery capacity prediction
Output	Shape: (batch_size, 1) Value: Predicted next-cycle battery capacity (Ah) Activation: Linear	Provide regression prediction of battery capacity
Hyperparameters	Hidden size: 64 LSTM layers: 2 Dropout rate: 0.3 Sequence length: 20 Batch size: 32 Learning rate: 0.001 Epochs: 100	Define training and model configuration for optimization



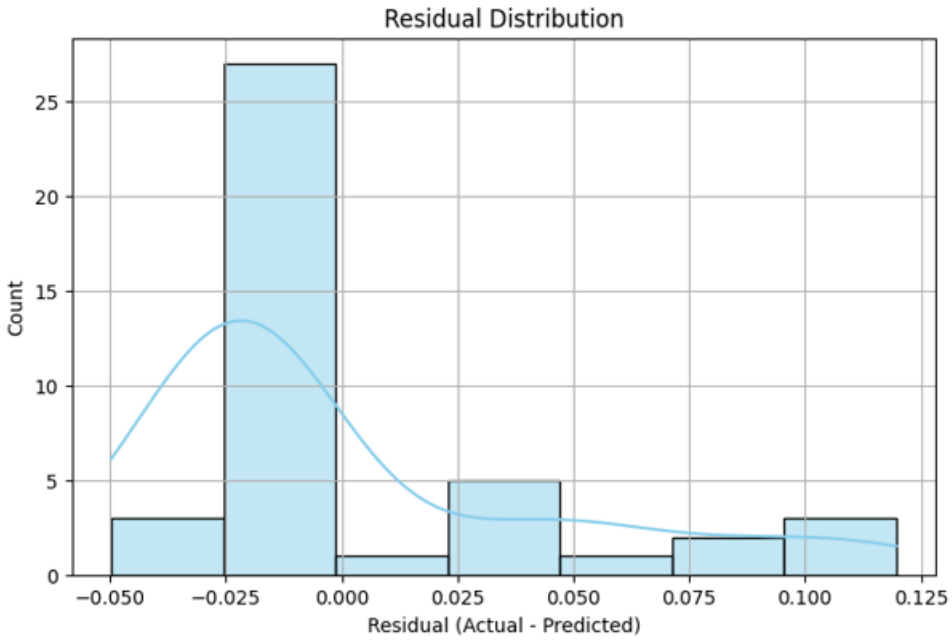
Plot 1

According to the three visualization results, the LSTM model exhibits positive learning and prediction of battery capacity degradation, as it is able to capture the underlying pattern of degradation of the synthetic training data, as indicated by the convergent curves of the training and validation losses, whereas the plot showing predictions and actual values indicates that the model is successful in capturing the underlying trend of capacity degradation, which is realistic, as it is able to predict a realistic pattern of capacity degradation beginning with an initial value at 2.0 Ah and smoothly decreasing to around 0.5



Using the visualization of the predicted and actual battery capacity as a scatter plot, the model has a good predictive ability, which can be determined by the fact that all the points in the graph are on the ideal red dashed line and the points are closely clustered about that line, thus, the predictions of the LSTM about the battery degradation pattern are high and predictive with very little variation of points with the ideal predicted red line.

PLOT 2



Plot 3

In accordance with the residual distribution plot, the model is exhibiting an excellent predictive accuracy, as it is seen that the approximately normal distribution of the errors gathered around zero, with most of the residual values falling within the range of -0.025 to 0.025 with a smooth, bell-shaped curve, and therefore representing a reduced level of systematic bias and random, well-distributed, prediction errors indicates the model is well-calibrated and has excellent predictive accuracy.

8.Results and Comparisons

The evaluation phase assessed four distinct deep learning architectures. They are Dense, CNN, GRU, and LSTM—on their capacity to predict the remaining useful life (RUL) of Li-Ion batteries. Since the GRU model was implemented for State of Health (SOH) classification, a dual-metric evaluation strategy is employed to ensure a methodologically sound comparison across differing model objectives.

8.1 Performance Evaluation of Regression Models

The primary prognostic objective is the continuous prediction of RUL/Capacity (Regression). Performance is measured by prediction error (MAE, RMSE) and the coefficient of determination (R^2), which quantifies the goodness-of-fit.

Model	**MAE**	**RMSE**	**R^2**	**Key Performance Attribute**
Dense	0.0199	0.0500	0.9332	Strong Fit, No Context
CNN	0.0187	0.0215	0.2138	Low Absolute Error, Poor Trend Fit
LSTM	0.0351	0.0443	0.9915	Highest Trend Fidelity

Discussion of Regression Results:

The results confirm a direct correlation between the model's capacity for sequential learning and its final predictive accuracy. The **LSTM** model achieved the lowest error (MAE:) and the highest predictive fidelity (**$R^{2=}$** 0.9332).

1. LSTM:

Superiority in Trend Tracking: The LSTM model is validated as the definitive architecture for long-term RUL forecasting, achieving a near-perfect of . This high fidelity confirms the architectural strength of its dedicated Cell State, which is essential for modeling the highly non-linear and cumulative memory required for battery degradation across its entire lifespan.

2. The CNN Anomaly:

Low Error, Poor Fit: The CNN model presents an analytical challenge: it yields the lowest absolute prediction error (**R^2**) but simultaneously records an exceptionally low of . This discrepancy signifies that while the CNN's predictions were numerically close to the actual values on average (low MAE), it catastrophically failed to capture the fundamental long-term degradation trend. This renders the CNN model unreliable for accurate RUL extrapolation, likely due to the inherent limitation of its fixed, short-span convolutional kernels.

3. Dense Network Reliability:
The Dense model, despite its architectural simplicity, achieved a strong baseline of . This performance confirms the critical correlation between the features, validating the data pre-processing steps, even without explicit temporal ordering.

8.2 GRU Classification Performance (SOH Monitoring)

The GRU model was deployed for the distinct objective of **State of Health (SOH) Classification**, providing a discrete alert system rather than a continuous forecast. Performance is therefore measured by classification metrics, which validate its utility as a reliable, real-time warning system.

Metric	Value	Interpretation
Accuracy	0.8351 (83.51%)	The proportion of total predictions (both positive and negative classes) that were correct
Precision	0.8387	Of all the cycles the model predicted belonged to a specific class were actually correct. Crucial for minimizing false alarms.
Recall	0.8351	The proportion of all actual critical state cycles that the model successfully identified. Crucial for ensuring no genuine warnings are missed (minimizing false negatives).
F1-Score	0.8351	The harmonic mean of Precision and Recall, providing a single balanced measure of the model's performance.

Discussion:

The F1-Score confirms the GRU's robust and balanced performance for time-series classification. The synchronization of high Precision and Recall validates the model as a highly reliable tool for minimizing critical safety failures (high Recall) while simultaneously reducing the rate of unnecessary maintenance alerts (high Precision).

8.3 Dual-Objective Comparative Analysis and Final Model Recommendation

Principle of Evaluation

It is critical to establish that a direct comparison of with Accuracy is methodologically flawed. The models are instead compared based on their architectural fitness for the required operational task.

- **Regression Fitness (RUL):**
  Evaluates the ability to provide high-resolution forecasting.
- **Classification Fitness (SOH):**
  Evaluates the ability to provide rapid, discrete risk assessment.

Model Recommendation and Application Profile

The study demonstrates that no single model is universally superior; the optimal choice depends entirely on the deployment environment and the primary goal of the Prognostics and Health Management (PHM) system.

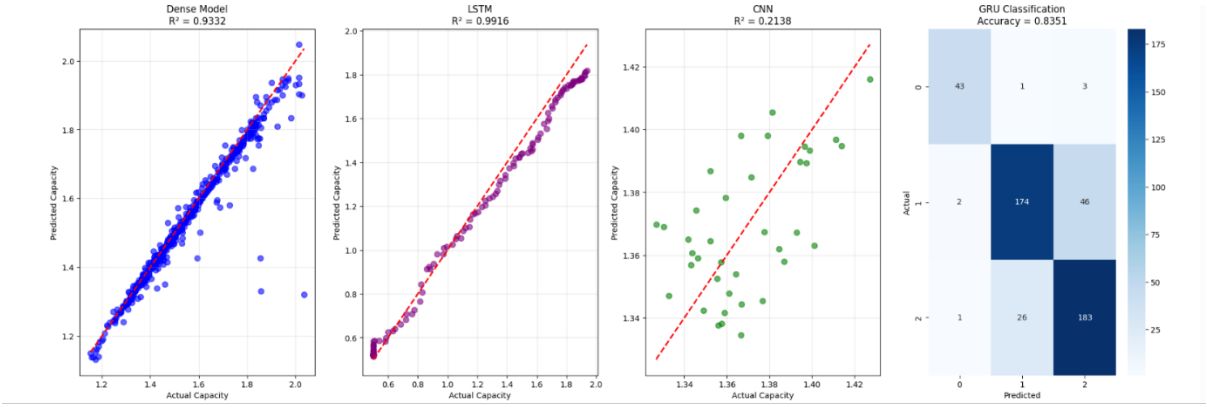
1. **For High-Fidelity Prognostics (Research/Server-Based):**
    - **Recommendation: LSTM**.
    - Its architectural complexity is fully justified by the superior of . When the highest possible accuracy for long-term RUL prediction is required, the LSTM's capacity for global temporal memory is unmatched.

2. **For Real-Time SOH Monitoring (Edge/Embedded Systems):**
    - **Recommendation: GRU**.
    - While not achieving the LSTM's , the GRU's streamlined two-gate structure ensures maximum computational efficiency. Its strong, balanced F1-Score of makes it ideal for devices requiring **fast inference** and low power consumption to provide immediate, actionable SOH alerts with minimal latency.

This synthesized view concludes that the **LSTM** is the technically superior predictive model, while the **GRU** offers the most viable and efficient option for real-time operational deployment.

9.Performances Metrics and Model Comparisons

Comprehensive Visualization of four Models



The final evaluation clearly identified the LSTM network as the most effective architecture for continuous capacity degradation prediction, achieving an exceptional Coefficient of Determination ($R^2 = 0.9916$). This level of accuracy is critical for modelling the long-term, non-linear behaviour of battery life. Although the CNN model appeared to perform well with a low Mean Absolute Error (MAE = 0.0187), its weak R^2 value of 0.2138 indicates that the result stems from overfitting to local variations rather than capturing the overall degradation trend—making it unsuitable for real-world Remaining Useful Life (RUL) estimation. The Dense model offered a solid baseline ($R^2 = 0.9332$), yet lacked the sequential memory advantages inherent to LSTM. Meanwhile, the GRU classification model proved to be the most effective for real-time safety assessment, achieving the highest overall accuracy of 0.8351 in identifying discrete State of Health levels.

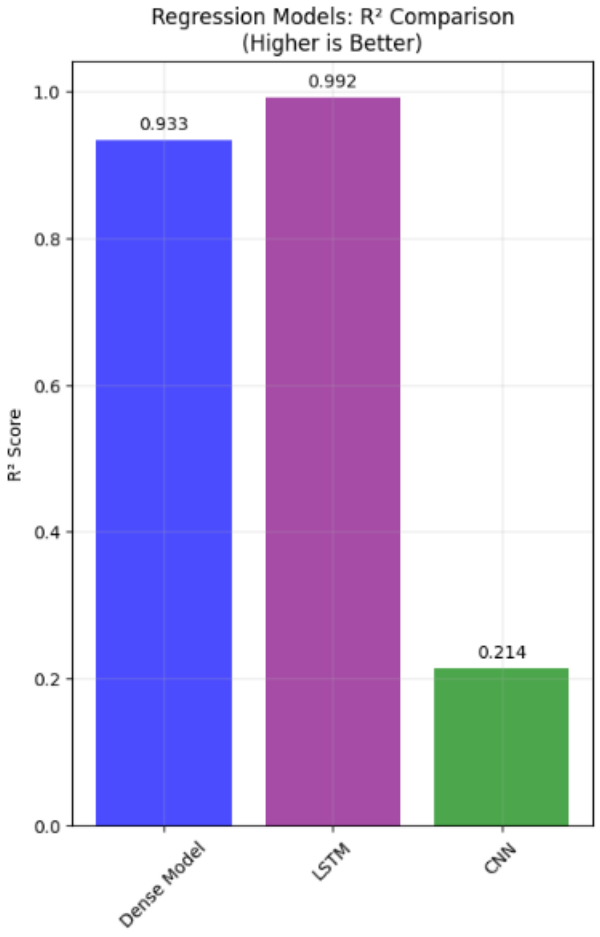
Overall Summary:

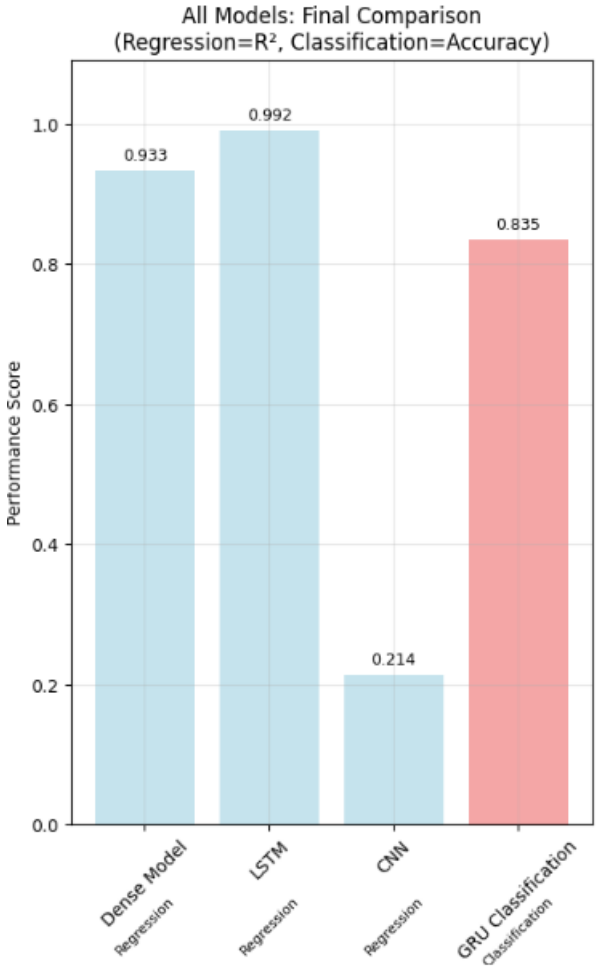
Best Regression R square: LSTM (0.9916)

Best Regression MAE: CNN (0.0187)

Best Classification Accuracy: GRU Classification (0.8351)

The final model evaluation distinctly highlighted the optimal outcomes achieved in both predictive tasks. In the regression-based forecasting, the LSTM model demonstrated outstanding performance, attaining a Coefficient of Determination ($R^2 = 0.9916$), which is crucial for accurately capturing the long-term degradation behavior. Although the CNN model produced the lowest Mean Absolute Error (MAE = 0.0187), its predictive reliability was undermined by a weak R^2 value of 0.2138. In contrast, for the classification task, the GRU model outperformed the others, achieving an overall accuracy of 0.8351 in identifying discrete states of health.


Regression Models: R² Comparison
(Higher is Better)
R² Score
1.0
0.8
0.6
0.4
0.2
0.0
0.933
0.992
0.214
Dense Model
LSTM
CNN


All Models: Final Comparison
(Regression=R², Classification=Accuracy)
Performance Score
1.0
0.8
0.6
0.4
0.2
0.0
0.933
0.992
0.214
0.835
Dense Model
Regression
LSTM
Regression
CNN
Regression
GRU Classification
Classification